

The medicine is already there. Nobody can see it.

A federated early-warning grid for India's primary health network. Demand is forecast at every facility–drug position — on Google's **TimesFM** wherever a held-out backtest says it wins — the detector spots an outbreak surge days before a shelf empties, and the planner moves medicine across district lines before it does, without creating a stock-out anywhere else.

Build with AI: Code for Communities

Problem statement 03 – Smart Health & Supply Chain Resilience

Live: `aaogya-grid-215071922486.asia-south1.run.app`

12,010

FACILITIES TRACKED

3,53,558

STOCK POSITIONS

23,070

DISPATCH ORDERS PLANNED

36

FEDERATED STATE NODES

THE PROBLEM

A sub-centre runs out of anti-snake venom in monsoon. The vials exist.

They are ninety minutes away, in a facility that will write them off unused in six weeks. Nobody knows either fact.

Both facilities report into a paper register that reaches the district office weeks later, if at all. India's public health supply chain does not primarily fail on procurement volume — it fails on **visibility**, and on **lateral movement**: the ability to see a shortage forming and move stock sideways before it becomes a stock-out.

20,511

POSITIONS AT ZERO STOCK

16,772

CRITICAL POSITIONS

121.12 Cr

PEOPLE IN CATCHMENT

₹58.61 L

STOCK HEADING TO EXPIRY

769 districts across all 36 states and union territories · zero-stock and critical are as-of positions on 2026-09-30 · expiry is the next 90 days · facility-level stock is simulated, see slide 13

Four users, one dataset — and a loop that comes back.

ANM SUB-CENTRE	Speaks one sentence into a phone instead of filling a register. Told at once if what she reported looks wrong.
Storekeeper PHC / CHC	Gets an order naming a batch, a quantity, an expiry and a distance. Nothing to interpret.
District Health Officer DISTRICT	Asks in their own language, gets a grounded answer with its working shown, and signs what can move.
State & ministry NATIONAL	Sees which districts fail structurally rather than unluckily — and gets a national demand model without any state handing over its facility rows.

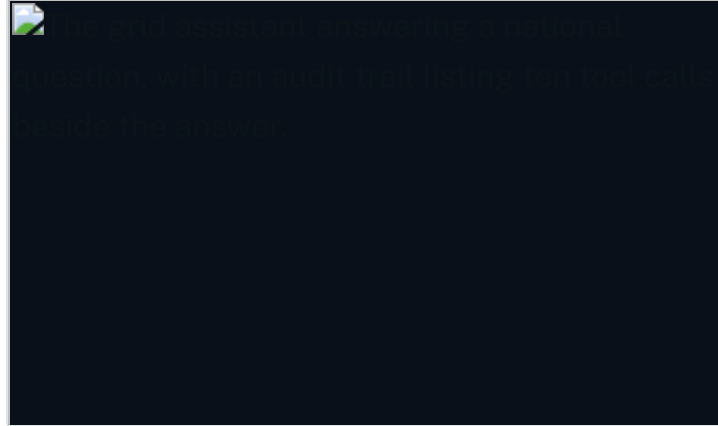
- OBSERVE** Gemini transcribes spoken Hindi or reads a photographed register. **A human confirms** before anything is committed.
- PREDICT** TimesFM forecasts 21 days ahead; a Monte Carlo over the lead time turns that into a stock-out probability and an expected shortfall.
- RECOMMEND** A dispatch order: named batch, expiry date, distance, price — bounded by what the batch physically holds.
- EXECUTE** approve → dispatch → receive, on an append-only audit trail. A short receipt is a **variance**, not a rounding.
- MONITOR** Re-scored server-side in **14 ms** and pushed over SSE. Survives a reload and a killed container.

Most systems of this shape stop at **Recommend**. Nothing afterwards comes back, so nobody learns that 14 vials were sent and **11 arrived**. Here the receiver's risk recovers by what turned up.

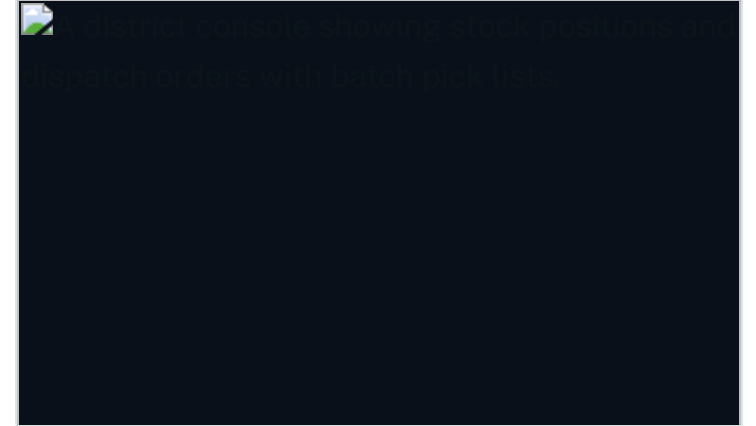
The national view, the assistant's working, and one district's orders.



1 — **/console**. Every bubble is a district; every arc is medicine actually moving.



2 — **Ask the grid**. The audit trail is the product: ten tool calls, every row they read.



3 — **/district**. Batch, expiry, price — and who has to countersign.

Not a dashboard alert. A dispatch order a storekeeper can act on.

GENERATED DISPATCH ORDER · SOUTH WEST KHASI HILLS → WEST KHASI HILLS, MEGHALAYA

SC West Khasi Hills-02 holds **1 sachet** of oral rehydration salts — a **Vital** drug — against a **100%** chance of running short inside its 15-day resupply window. **CHC South West Khasi Hills-01**, in the next district, can spare **107**. Moving them **40.2 km** costs **₹1,174** and averts an expected **86.0 sachets** of unmet demand, cutting stock-out risk from **100% to 4%**.

pick list: B012-HC001 × 23 (exp 2028-11-07) · B013-HC001 × 84 (exp 2029-04-29) · risk reduction 96 pp · donor left at 0% stock-out risk

Every recommendation names the **specific batches and their expiry dates**, oldest first, and the quantity is bounded by what those batches physically hold. A recommendation a storekeeper cannot fill is not a recommendation. This one crosses a district boundary — the clause the brief adds this edition — so it also carries who is allowed to issue it: **West Khasi Hills cannot approve it alone**, and the console disables Approve until South West Khasi Hills countersigns. This order pays for its own vehicle — **₹1,174** planned, **₹1,174** alone — and that vehicle then carries **3 other orders** that ride along for the price of handling.

TimesFM forecasts. Gemini reads and retrieves. Neither invents a number.

- TIMESFM 2.0

All **36,143** district × drug series, **21 days** ahead from a **90-day** context, through BigQuery **AI.FORECAST** — 13 concurrent statements, **0 bytes billed**, none declined.
- GEMINI

Spoken Hindi and photographed registers into structured entries; and a **function-calling** surface of **12 tools** over the computed state, with the **tool trace on screen**.
- DETERMINISTIC

Censored Croston-SBA, Monte Carlo risk over each facility's own **7–18 day resupply window**, and the redistribution optimiser. Reproducible from a seed.
- NEVER THE MODEL

No arithmetic, no drug code, no facility id, no status. The drug is resolved **server-side from its name**, never from a client-supplied id.
- CENSORING

A ledger records what was **dispensed**, not what was **needed** — a stock-out looks exactly like a quiet week. Fitting naively under-orders for the facilities already failing, so the fit excludes stocked-out periods.

DEMAND CLASS	POSITIONS	TIMESFM	CROSTON	SERVES
Intermittent	98,844	0.894	0.968	TimesFM
Smooth	1,14,566	0.763	0.771	Croston
Erratic	69,163	0.711	0.701	Croston
Lumpy	32,699	0.807	0.845	Croston

MASE on a **28-day held-out** backtest neither model saw. TimesFM takes a class only where it beats Croston by more than **5%**. It wins intermittent demand by 7.7% and holds **1,36,331** of **3,53,558** shipped positions. **We publish where it loses.**

AND THE FIT IS ON CORRECTED HISTORY	BIAS	ERROR
The raw ledger	-3.2%	3.4%
Stock-outs excluded	-0.9%	2.0%

Never MAPE: it divides by the actual, and intermittent demand is full of zeros · full table in docs/forecast-backtest.md · censoring: 3,871 pairs, error = mean |fitted - truth| / truth, src/data/censoring-eval.json

A stock-out is a lagging indicator. We publish how often the warning is wrong.

By the time a block's anti-malarials run short, the malaria has been there a fortnight. **AI.DETECT_ANOMALIES** runs over district consumption and OPD attendance — 36,912 series, 32 statements, **0 bytes billed** — and a warning is a **rule over points**, tuned against **126 injected surges** and scored on 80 candidate rules.

Detection of a 2× 14-day surge	100%
Median lead before the first shelf empties	4.58 days
False alarms per district-week	0.446
Precision	21%

21% precision is not a good number and it is on this slide anyway, next to the 78 rules that failed. Two tighter rules reach 53% and 42% and miss only the four-day lead. Moving the gate after seeing the table would turn every number here into an argument.

OBSERVED · KERALA IDSP DAILY BULLETINS

242 bulletins read from their text layer, every column checked against its printed total. The same rule flags **Kasaragod fever consultations, 10–11 Sep: 2,485** against at most 1,932 — each signal linked to the bulletin it came from.

SIMULATE_OUTBREAK · PURNIA, VECTOR-BORNE ×2

6 of 78 surge needs servable at routine valuation, **17 at emergency**, for **₹14,867** more transport. An outbreak has to raise the **price of a stock-out**, not only demand — a ministry dial, reported next to its effect rather than baked in.

States keep their data and share only models. Here is exactly what crossed.

56,660

NUMBERS SHARED, 36 NODES

65,05,740

ROWS THAT STAYED PUT

- facility rows
- stock quantities
- patient records
- district identifiers

Each state fits its own model and publishes a monthly demand multiplier per catalogue item, its standard error, and one vacancy rate per cadre. Pooled by **random effects with the between-state variance estimated from the nodes**, so a state keeps its own estimate exactly to the extent its data earns it.

Checkable, not assertable. GET /api/federated/10 returns Bihar's node byte for byte; the index publishes the SHA-256 of all thirty-six. **npm test** sweeps every file for facility ids, district codes and batch numbers — after first proving on **five poisoned copies** that the sweep can still see one.

A STATE JOINS WITH 30 DAYS	SCALED MAE
No seasonality at all	0.2675
Its own fit, alone	0.2637
Shrunk toward the national prior	0.1687
Its own fit on all 180 days (ceiling)	0.1657

36.0% closer to observed demand than forecasting alone, measured **leave-one-state-out** — the prior it is offered is pooled from the other thirty-five only.

Limitation, stated before anyone finds it: one seeded simulator generates all thirty-six, so the pooling weights are a demonstration of a mechanism rather than a finding about Indian states. It is on the console panel too.

It does not pay for itself in cash. It pays in shortfall, and we lead with the sign.

NATIONAL PLAN	VALUE
Waste averted	+ ₹18.9 L
One dedicated vehicle per order	— ₹277.0 L
Transport cost, consolidated	— ₹125.9 L
Net cash position	— ₹107.0 L
Shortfall averted	34,39,003 units

It breaks even at ₹3.11 per averted unit of unmet demand. Whether a dose of a Vital medicine reaching a patient is worth that is a policy judgement, not an engineering one — so the shortage penalty is an explicit parameter, not something folded into a headline.

Visibility was not the binding constraint. A truck per tablet was. Of 34,576 needs the planner declined, 97.9% failed the benefit/cost gate — the medicine existed, a median 19.9 km away, and the trip cost more than the medicine. Only 8 failed for want of stock anywhere.

THE SAME 23,070 ORDERS	A VEHICLE EACH	CONSOLIDATED
Vehicle trips	23,070	9,421
Crossing a district	15,930 trips	6,415 trips

11,279 of those orders could not justify a vehicle alone. They ride a trip already going, for the price of handling — stock-outs that were being refused for a truck nobody needed to hire.

One region, every Google service named, nothing that is not a managed service.

FIELD

ANM speaks or photographs
phone browser

↓

Gemini 2.5 / 3.5 Flash
Vertex AI · asia-south1

transcription · translation · extraction · function calling

BATCH, NIGHTLY

Cloud Scheduler → Cloud Run Job
IDSP bulletins · seeded simulator

↓

BigQuery AI.FORECAST
TimesFM 2.0 · asia-south1

↓

BigQuery AI.DETECT_ANOMALIES
inline subquery · 0 bytes billed

↓

Cloud Storage runs/
769 plans, re-proved nightly

SERVE & REMEMBER

Cloud Run, 1–4 instances
Google sign-in · asia-south1

↓

BigQuery aarogya_grid
stock_events · dispatch_tickets

↓

Pub/Sub aarogya-events
every instance hears every commit

commit → re-score 14 ms → SSE → survives a restart

Every inference call stays in asia-south1 · the always-on instance is ~₹1,200-1,800/month and the nightly job is billed by the second · the forecast and detection queries process 0 bytes · commands: docs/operations.md

Stated plainly, because the honest answer is the stronger one.

REAL Notified cases in Kerala's **IDSP daily bulletins**, behind the observed signals · 769 districts at their headquarters towns, with LGD codes · **Census 2011 district populations**, reconciled to state totals · the NLEM drug catalogue, forms, strengths and VED classes · IPHS tier, bed and staffing norms · an NFHS-5 indicator for every state · the national boundary from DataMeet's composite file.

SIMULATED Facility-level stock, consumption, batches and expiries · bed occupancy · staff attendance · OPD footfall. Labelled **SIMULATED FACILITY DATA** in the console header and on the landing page, and itemised in NOTICE.

Why simulated? Because the absence of this data *is* the problem. If per-facility stock across India's PHC network were available as an extract, the shortage would already be visible. **Three adapter functions** stand between this and a DVDMS or e-Aushadhi extract; nothing downstream changes.

OWN IT Every write needs a Google sign-in and names who acted, but **roles are claimed**: there is no role directory to check them against.

OWN IT Two instances are **rehearsed, not load-tested**: they agree on every report and approve a raced order once.

OWN IT The federated τ^2 is **synthetic** — one simulator behind all thirty-six.

OWN IT The observed signals are **unvalidated**: the rule's precision was measured on simulated surges.

Nothing on any surface is typed by a human. A drift guard re-derives every figure in this deck, the README and the defence pack from the shipped payloads on every **npm test**. It has caught a slide quoting a dispatch order that no longer existed.

One state, eight weeks, two metrics — then the same code, more rows.

WEEK 1–2 Point the three adapters at one state's DVDMS/e-Aushadhi extract. Positioned as a **decision layer over** DVDMS, e-Aushadhi and IHIP — not a replacement.

WEEK 3–6 Two districts live: the ANM capture path, the DHO console, dispatch tickets in the district office.

WEEK 7–8 Measure **two** things and nothing else: **stock-out days prevented**, and **time from alert to action**.

REACH **2,348** district-to-district corridors carry medicine in the shipped plan, **284 cross-state corridors** among them — each one classified, and each one refusing to be approved by a district officer alone.

SCALE All 769 districts in **900–1,300 s** on one laptop, measured rather than extrapolated. District-parallel with no shared state; planning, which shares stock, colours into **10 rounds** of disjoint clusters, not 769 tasks.

BRICS portability. The core is country-agnostic: forecasting, censored fitting, risk, and a distance-and-price optimiser. What is country-specific is a **medicines list**, **facility norms**, a **currency** and a **language pack** — four adapters, not a rewrite.

The early-warning feed already leaves in a schema nobody has to be Indian to read: an area has a code, a named code system, a name and a population; a signal has a hazard class from a fixed list, an observed value, an expected range and a confidence. Every Indian identifier travels in an optional block a consumer can drop, and **npm test** strips that block and re-validates.

WHAT WE WOULD BUILD NEXT, IN ORDER

IDSP bulletins from a second state, and the national IHIP feed · a role directory, so the role on an audit row is checked rather than claimed · per-state backtests once two states have real history · and the thing a ministry will ask for first: **procurement forecasting**, which is the same demand model pointed a quarter further out.